Task 2 - Values Audit

There are no correct answers, but shallow answers would hide the trade-offs. This audit states them directly.

Company mission and identity

The hypothetical company is Meridian FairAI Governance, an early-stage RegTech firm that helps cross-border financial institutions govern AI credit models through explainable, auditable and jurisdiction-aware compliance tools. Its core values are practical fairness, transparent accountability, regulatory humility and human judgement. The company is deliberately small: about 25-40 specialists across model risk, AI explainability, compliance operations, regulatory engineering and customer implementation. Its near-term aspiration is a focused enterprise SaaS model for US/EU banks: low hundreds of thousands of dollars per annual bank deployment, implementation support sold separately, and expansion only where Meridian can keep legal humility and model-governance quality intact.

Whose perspective the tool serves

The paying users are the bank's CCO, CRO, Fair Lending Team, AI Governance Team and Model Risk Management function. That choice is partly commercial: these teams have budget, urgency and accountability. But the tool also embeds consumer and public-interest concerns by measuring approval gaps, false-positive / false-negative differences and adverse-action reason quality. There is a real tension: the bank may want efficient approvals and lower regulatory exposure, while consumers want fair access and understandable reasons, and regulators want market-wide accountability.

Task 2 - Stakeholder Trade-Off Matrix

Values audit: who benefits, who bears risk, and what the product does in response.

Stakeholder	What they want	If the tool is wrong	Design response
Citi Compliance / Model Risk	Audit evidence, explainable outputs, rule-version control	False comfort; wrong jurisdiction; missed regulatory issue	Versioned rules, conflict notes, monthly fairness review
Consumers / applicants	Fair access, understandable reasons, ability to challenge	Wrong denial, opaque explanation, unequal review burden	Specific adverse-action mapping; positive-risk SHAP features only
Regulators	Repeatable governance evidence, human accountability	Dashboard theatre without real risk detection	FPR/FNR/parity metrics plus limitations and escalation owner
Meridian	Defensible product niche and repeatable subscription model	Overclaiming legal compliance; loss of trust	Narrow scope: governance tool, not legal advice or auto-approval

Visual takeaway: the product is not neutral. It chooses to make trade-offs visible, documented and reviewable.

Task 2 - Values Audit

Genuine risk or documentation?

The tool is designed to measure genuine model and consumer harm risk, not merely to produce a checklist. One design choice that reflects this is group-level monitoring of approval parity plus false positive and false negative rates. These metrics are not required simply to produce a document; they help identify whether a model is systematically denying lower-risk applicants from a protected group or approving higher-risk applicants in a way that later harms the bank and consumers. Documentation remains necessary, but the dashboard starts from risk detection.

Who bears the cost when the tool is wrong?

A false positive can wrongly classify a low-risk applicant as high risk, causing a consumer to be denied credit or pushed into worse terms. A false negative can approve genuinely risky credit, increasing losses for Citi and possibly encouraging unsustainable borrowing. A fairness false negative is more serious from a public-interest perspective because it can miss systematic discrimination. A fairness false positive raises compliance cost and may slow lending without improving fairness. A jurisdiction misconfiguration can make the bank apply the wrong rule set, creating regulatory exposure and consumer harm. Model drift can silently degrade both credit performance and fairness over time.

Boundary of responsibility

The tool serves management judgement; it does not replace it. Its ethical stance is that automation should surface risks, evidence and unresolved tensions, while final credit policy, legal interpretation and customer remediation remain accountable human decisions.

Task 2 - Error-Rate Tension

The FNR result is not a simple fairness victory

The generated portfolio has a protected-group false negative rate of 0.3937 and a reference-group false negative rate of 0.4706. A lower FNR for the protected group means fewer actual defaulters in that group are incorrectly approved. On its own, that sounds safer. But the protected group also has a lower approval rate, so the model may be achieving lower default leakage partly by being more conservative for that group.

Who is helped and who may be harmed

From the bank's risk perspective, lower protected-group FNR reduces credit losses. From the consumer perspective, the same pattern can mean more borderline protected-group applicants are denied or pushed into manual review. The ethical tension is therefore between safety-and-soundness risk and access-to-credit harm.

How the tool should present it

The tool should not label the lower FNR as proof of fairness. It should show FNR together with approval parity, false positive rate and adverse-action reasons. If approval parity is weak while FNR is low, management should ask whether the model is too conservative for one group rather than simply celebrating lower default risk.

Task 2 - Model Choice as Value Choice

Random Forest creates a new ethical tension

The prototype includes a Random Forest challenger to test whether a more complex non-linear model materially improves performance. In this run, the challenger does not provide a significant advantage over logistic regression, while it is harder to translate into stable consumer-facing adverse-action reasons without additional SHAP validation.

Why logistic regression is not only a technical choice

Choosing logistic regression as the primary governance model is therefore a values decision as well as a modelling decision. Meridian is prioritizing explainability, auditability and the consumer's ability to understand or challenge a denial over a small or uncertain performance gain from a more opaque model.

Genuine tension

This choice is not cost-free. A more complex model could eventually capture legitimate non-linear risk patterns. But if that extra complexity does not clearly improve outcomes, using it in a credit context may shift too much burden onto applicants who receive less understandable reasons. The product's default position is that model performance must earn its opacity.

Task 2 - Commercial and Ethical Choice

Why this customer segment can pay

This product choice is primarily based on Meridian's competencies, not market size. The team's advantage is explainable AI, model-risk governance and jurisdiction rule configuration, so AI credit governance is where it can build a defensible product. The market evidence supports feasibility rather than driving the decision: Grand View Research estimated the global RegTech market at USD 24.34 billion in 2025 and projected USD 112.10 billion by 2033, with large enterprises and risk/compliance management as important segments. A global bank therefore has both the compliance budget and the operational need; Meridian's constraint is not demand, but staying narrow enough that its controls remain auditable.

Tension I am not hiding

A bank may buy this tool to reduce regulatory exposure, while a consumer may want a simpler answer: was I treated fairly and can I challenge the decision? The tool tries to reduce that gap by forcing specific adverse-action reasons and group-level harm metrics into the management view. It still cannot guarantee that the consumer understands the explanation or has equal bargaining power.

A feature intentionally not included

The prototype does not auto-retrain the model when a fairness flag appears. Automatic remediation would look sophisticated, but it could hide policy choices inside code. The tool instead escalates to human review because deciding whether to change a credit model, adjust a threshold or remediate customers is a governance judgement, not a purely technical optimization.

Task 2 - Evidence Boundary

Case-study honesty

The Citi evidence supports a realistic cross-border banking case, but it does not prove Citi's actual EU AI-credit exposure. Meridian would treat a real sales cycle as discovery-led: first verify model inventory, jurisdictions, products, customer segments and decision workflows, then configure the governance controls. This avoids turning public research into unsupported claims about a named bank.

Why this still fits the assignment

The assignment asks for a named regulated entity and a RegTech product opportunity. The submission therefore uses public evidence to justify Citi as a plausible cross-border case and uses synthetic data to demonstrate the product mechanics. It does not claim access to proprietary Citi model documentation.

Task 2 - Competitive Position

What Meridian is good at

Meridian's competence is not generic consulting. Its niche is the translation layer between model outputs, explainability evidence, fairness metrics and jurisdiction-specific governance actions. The product is designed for model-risk and compliance teams that need repeatable evidence rather than one-off memos.

Differentiation from larger incumbents

Large consultancies such as Accenture or KPMG can advise on regulatory transformation, and established RegTech vendors can support KYC, AML or entity resolution. Meridian's differentiator is narrower: AI credit governance that connects SHAP-style explanations, adverse-action reason evidence, fairness thresholds and rule versioning in one workflow. The company wins by being more specialized and auditable, not by offering every compliance service.

Revenue logic

The likely commercial model is annual subscription plus implementation support for large banks, with pricing tied to number of models, jurisdictions and governance users. This aligns with the buyer's ability to pay because the tool supports recurring model monitoring, rule updates and audit evidence rather than a one-time report.